1. Introduction

Deep learning has revolutionized computer vision in recent years. Convolutional neural networks have achieved state-of-the-art results on many benchmark datasets including ImageNet and COCO.

However, current approaches still face significant challenges. Domain shift remains a major obstacle to real-world deployment.

2. Method

We propose a novel architecture called Residual Attention Network. Our method combines skip connections with channel-wise attention mechanisms to improve feature representation.

The key insight is that different channels encode different semantic information, which should be weighted differently.

3. Experiments

We evaluate on ImageNet, CIFAR-100, and fine-grained datasets. Results show 2.3% improvement over ResNet-50 baseline. Ablation studies confirm the effectiveness of attention modules.